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CHAPTER

Catalyzing Intrinsic and Internalized Motivation: The Role of Basic Psychological Needs and Their Support

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Abstract

According to Self-Determination Theory, intrinsic motivation is a natural growth tendency that fuels students' learning, development, and well-being. In this chapter, we sketch a developmental model of intrinsic motivation, thereby describing an evolution from fleeting and undifferentiated interests to the gradual unfolding of an enduring, refined, and well-anchored intrinsically motivated identity. To foster such mature forms of intrinsic motivation, learners' psychological needs for autonomy, competence, and relatedness need to be supported on both the short and long run. Yet, because not all learning material and school-related activities are interesting, a critical task for socializing figures is to foster the growth process of internalization, such that learners experience a sense of ownership regarding extrinsically motivated activities. Teachers and school climates that are conducive to learners' psychological needs promote learners' endorsement of the learning activity at hand, while also allowing them to ground their functioning on a deeper set of values, commitments, and preferences, which serve as a more stable basis for regulating their day-to-day learning activities. Apart from discussing these growth processes and the energizing role of psychological needs, we review past work which provides insights in the way how teachers can foster learners' intrinsic motivation and internalization.

Keywords: self-determination theory, psychological needs, intrinsic motivation, internalization, identity development

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When observing toddlers or preschoolers in a classroom, one can easily notice children being highly enthusiastic and eager to learn. Children's curiosity leads them to get fully absorbed in the activity at hand and to be spontaneously interested in the content of the activity and learning material itself. From the perspective of Self-Determination Theory (SDT; Ryan & Deci, 2017; Ryan et al., 2021), these children display intrinsic motivation, a process that manifests through acts of exploration, manipulation, and discovery. According to SDT, intrinsic motivation is part of our active human nature and fuels students' learning, development and well-being.

Yet, because not all learning material at school is interesting—at least not all the time and not for all learners—an important question is whether other natural sources of motivation may play a supplementary role (in addition to intrinsic motivation). To foster learning and persistence even when learning activities are hard (i.e., dull or difficult), it is important for students to see the personal relevance of such activities (Ryan & Connell, 1989; Vansteenkiste et al., 2018). Through a process of increasing internalization, children then learn to take responsibility for their learning and behavior even when the learning activities are not intrinsically rewarding.

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Although intrinsic and internalized motivation constitute natural growth processes, for children to discover, maintain, refine and harmonize their interests, values, and commitments and for them to fully endorse their learning activities, ongoing contextual support is required. High-quality motivation will be catalyzed when students find themselves in an environment supportive of their psychological needs for autonomy, competence, and relatedness or, alternatively, when they manage to steer their own development toward elevated need satisfaction themselves (Vansteenkiste et al., 2020). In this chapter, adopting a SDT perspective, we discuss several critical features of a developmental motivation model and selectively review past research highlighting the role of psychological needs in the development and maintenance of both intrinsic motivation and internalization.

Intrinsic Motivation

We begin with conceptualizing the growth process of intrinsic motivation and its development. After having contrasted intrinsic motivation with a hedonic orientation, we review past work, which shows that intrinsic motivation constitutes a natural resource of learning and development.

The Development of Intrinsic Motivation

The process of intrinsic motivation is central to *Cognitive Evaluation Theory*, the mini-theory that formed the starting point for the development of SDT half a century ago (Deci & Ryan, 1985; Vansteenkiste et al., 2010). When intrinsically motivated, learners engage in a learning activity for its own sake. *Intrinsic motivation refers to engaging in a school activity because it is perceived as interesting, enjoyable, and challenging in its own right* (Ryan & Deci, 2017, 2020). Not only learning activities, but any school-related activity can vary in its level of intrinsic motivation, including children's free play during the midday break or their engagement in extra-muros activities. Intrinsically motivated activities are autotelic in nature because the goal of the activity lies in its engagement as such and the satisfaction is inherent to the activity itself. To illustrate, a child may like reading comics or novels (De Naeghel et al., 2012), love to play music (Evans & Ryan, 2022), or feel challenged to solve math exercises (Guay & Bureau, 2018).

In contrast, when extrinsically motivated, learners engage in an activity to attain an outcome separable from the activity itself. Extrinsic and instrumental motivation are conceptually synonymous as the learning is a means to an end or an instrument to achieve a separate outcome. A classic example of extrinsic motivation is children engaging in the activity to obtain contingent rewards (Deci et al., 1999). Yet, learners' activity engagement can serve a much broader set of outcomes, with some of these outcomes being positively valanced (e.g., the approach of approval, rewards, and values) and with other outcomes being negatively valanced (e.g., the avoidance of punishment, guilt, and regret).

Children's interests and preferences develop gradually throughout childhood. At a younger age, children get to know their interests through free play, discovery, and exploration of the environment. Their emerging interests play an *orienting* role, thereby leading them to select preferred activities. When supported ongoingly, these emerging "raw" inclinations, which are initially broad and largely implicit, develop more refined and explicit interests to which one is highly committed. For example, children initially attracted to construction play may gradually develop an interest into more specific types of construction play (for instance, with a specific preference for Lego, and for building medieval castles in particular). Later in life, these more refined and explicit interests may guide study career choices (e.g., STEM, archaeology) aligned with these initial inclinations. With age and under supportive circumstances, children's natural interests become more differentiated and crystallize into more enduring interests (Hidi & Renninger, 2006; Renninger & Hidi, 2011) that eventually come to define their identity (Ryan & Deci, 2003; Soenens & Vansteenkiste, 2011).

When intrinsically motivated activities begin to form the basis for individuals' identity, they have often developed into a *passion* (Vallerand et al., 2003). In the case of passion, children spend considerable time in an activity which they both value and find truly interesting, enjoyable, and even fascinating to carry out. When learners are passionate about a topic, they do not merely like the learning activity (e.g., math exercises), but they also define and present themselves to the outside world as a person holding these enduring interests (e.g., "I am an architect student"). That is, their identity is centered around these core interests that provide a sense of self-confidence and that form a basis for stable self-regulation. To

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illustrate, in a longitudinal study spanning 10 years, children whose personal identity already included a long-term perspective to become a music player at an early age were more likely to sustain instrument learning over time, an effect that emerged above and beyond their amount of practicing (Evans & McPherson, 2015).

Although students' intrinsic motivation is often assessed through self-reports, it also manifests through a number of interrelated emotional and behavioral indicators. At the emotional level, intrinsically motivated individuals often report feeling vitalized and energetic (Ryan & Frederick, 1997), and they display observable signs of enthusiasm (Kunter et al., 2011) and emotional engagement (Skinner et al., 2008). At the behavioral level, intrinsically motivated individuals typically continue engaging in the requested activity, even if they are no longer surveilled or rewarded (Deci et al., 1999). Because their interest lies in the activity itself, this is a sufficient basis for voluntary persistence, even in the absence of external surveillance. To estimate whether learners are intrinsically motivated, teachers may rely on these more observable manifestations as their ratings of learners' interest and enjoyment as such match rather poorly with learners' own self-reports of intrinsic motivation (Lee & Reeve, 2012; Ulstad et al., 2019).

Intrinsic Motivation and a Hedonic Orientation

A criticism sometimes voiced against intrinsic motivation is that its cultivation involved is based upon a rather naive or unrealistic model of educational practice. Intrinsically motivated learners would engage only in "fun" activities at school, would limit themselves to learning only the interesting parts of their course material, and may give up their schoolwork when alternative and more attractive leisure-time activities are looming. The caricature that intrinsically motivated learners would be selective and too easygoing in their efforts is based on the (mistaken) assumption that intrinsic motivation involves a *hedonic orientation* (Zimbardo & Boyd, 1999). Although both hedonically oriented and intrinsically motivated learners are driven by enjoyment and focus on the presence, they function very differently. Hedonistic learners are "locked" in the present because they are looking for immediate satisfaction and hedonic pleasures (e.g., partying, screen time, etc.). In the hedonic approach, well-being is equated with the pursuit of pleasure and happiness and denotes an immediate quest for gratification (Ryan & Deci, 2001). Yet, within SDT, intrinsic motivation reflects a growth-oriented tendency; it denotes the natural tendency to discover and enact one's emerging interests such that one can realize oneself, enabling one to experience a sense of eudaimonic well-being over time.

Interestingly, a hedonic orientation and intrinsic motivation to learn were even found to be slightly negatively correlated (De Bilde et al., 2011). Similarly, a hedonic orientation relates to less school interest and satisfaction (Braaten et al., 2019) and higher external pressure (Behzadnia & Ryan, 2018). Students may adopt a "carpe diem" attitude characteristic of present hedonism to overcome the lack of eudaimonic and, hence, self-actualizing interest they experience from their learning activities. In contrast to students high on intrinsic motivation, who often go the extra mile to learn more, learners with a hedonic orientation adopt a rather minimal approach to their schoolwork, thereby limiting themselves to what is strictly needed. They are more inclined to escape harder situations because they are low on energy (Zimbardo & Boyd, 1999).

Intrinsically motivated learners have more energy available because there is strong concordance between their learning activities and their deepest interests and preferences. With intrinsic motivation, learners feel that they can function authentically and be entirely themselves in the activity. They are fully immersed into their activities, are more likely to forget about time and space, and fully energized to engage in the activity at hand (e.g., Valenzuela et al., 2018). This energy can then be used to cope more efficiently with obstacles encountered during the learning process, such as when learners face harder or duller learning material or when alternative extracurricular activities risk drawing their attention. Congruent with the assumed growth-oriented nature of intrinsic motivation, intrinsically motivated learners reported using more adaptive coping when encountering obstacles (Ntoumanis et al, 2009) and were less likely to drop out of school (Renadu-Dube et al., 2015; Vallerand & Bissonnette, 1992). So, the idea that intrinsic motivation would come with reduced self-control to persist under difficult circumstances is not confirmed by research, even on the contrary. The engagement in intrinsically motivated activities allows learners to move along the eudaimonic path of increasing self-realization, thereby furnishing the necessary energy and determination to overcome problems.

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A Natural Resource of Learning and Development

Meta-analyses summarizing hundreds of studies across the world have shown that intrinsic motivation comes with a host of affective, cognitive, and behavioral benefits (e.g., Cerasoli et al., 2014; Howard et al., 2020; Vasconcellos et al., 2020). To illustrate, students with higher intrinsic motivation report more positive emotions and higher well-being (Ketonen et al., 2018); they process the learning material more deeply and obtain better grades (Cerasoli et al., 2014; Taylor et al., 2014); they are better in self-regulating their school work, as indexed by better planning and organization of their learning activities and less procrastination (Vansteenkiste et al., 2009); and they proactively seek and select more challenging activities (Mabbe et al., 2018).

The benefits associated with intrinsic motivation have been demonstrated in multiple age groups, ranging from toddlers (Hepach, et al., 2019), to elementary school children (De Naeghel et al., 2012), high school students (Howard et al., 2021), and college and university populations (Yu & Levesque-Bristol, 2020). Intrinsic motivation also relates to a host of desirable outcomes among teachers, including an increased use of more constructive and motivating teaching practices (Aelterman et al., 2019), higher teacher well-being (Abos et al., 2018), and decreased disruptive classroom behavior and symptoms of burn-out (Benita et al., 2019). The benefits associated with intrinsic motivation have been shown not only in correlational research but also in experimental research. The experimental priming of intrinsic motivation was found to enhance students' efforts and performance (Radel et al., 2013). For instance, elementary school children showed greater conceptual learning (Grolnick & Ryan, 1987) and more positive attitudes toward homework (Froiland, 2011) when their intrinsic motivation for school was boosted through experimental manipulations. Overall, it is safe to conclude that intrinsic motivation goes along with a smoother (i.e., less effortful) and more effective development of knowledge, skills, and attitudes (Mouratidis et al., 2017).

Although intrinsic motivation is a natural source of motivation and although children feel naturally attracted to engage in these activities under supportive circumstances, research across the globe paints a rather alarming picture of the prevalence and development of learners' intrinsic motivation over time. Rather unfortunately, children's intrinsic motivation declines substantially as they progress through primary and secondary school (Gottfried et al., 2001; Lepper et al., 2005; Nashimura & Sakurai, 2017; Wigfield et al., 1997). Longitudinal studies, spanning a single school year (Corpus et al., 2009) or multiple school years (Gottfried et al., 2009; Vedder-Weiss & Fortus, 2012), have provided evidence for this downward trend. Yet, it should be noted that not all students are equally vulnerable to this decline (Amiot et al., 2008) and that this trend was found to be more pronounced for some courses (e.g., math, sciences, reading motivation) than for others (e.g., physical education; Gottfried et al., 2001; Jaakkola, et al., 2015). For instance, Guay and Bureau (2018) showed such subject-specific patterns in students' school motivation as a function of age. Also, this decline over time is halted among those who embark studies in higher education, presumably because the chosen major fits with students' interests and preferences (Kyndt et al., 2015).

Catalyzing Growth: Basic Psychological Need Satisfaction

We begin by describing the notion of basic psychological needs and their functional role. We then review past work showing that situational need support is critical to foster intrinsic motivation, while enduring need support is vital to anchor intrinsic motivation. Finally, we discuss how the satisfaction of basic needs yields manifold additional benefits, including children's engagement, learning, and well-being.

The Role of Basic Need Satisfactions

The erosion of intrinsic motivation as a natural source of motivation raises the question how these trends can be buffered or even altered. Although intrinsic motivation constitutes a natural growth process, its development does not take place automatically but requires contextual supports. The maintenance and nurturance of intrinsic motivation is said to depend largely upon the satisfaction of students' psychological needs for autonomy, competence, and relatedness (Ryan & Deci, 2017). The study of these needs, which has received widespread attention during the last decade, is central to *Basic Psychological Need Theory*, another mini-theory within SDT (Vansteenkiste et al., 2020).

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Autonomy refers to the experience of psychological freedom and volition. When satisfied, individuals have the feeling that they can be themselves and that they are the "owner" of their thoughts, actions, and feelings. Students feel that there is room to make meaningful choices and that their viewpoint is heard by others (including their teacher). Competence denotes the experience of effectiveness and mastery. When satisfied, students feel confident and capable to execute a learning task, to attain personally valued goals, and to exercise and extend their skills. Relatedness denotes the experience of warmth, bonding, and mutual care. When satisfied, students feel strongly connected to their classmates and teacher (Klassen et al., 2012), while also displaying a reciprocal concern toward them.

Within SDT, it is maintained that these three psychological needs form an integral part of individuals' organismic functioning. That is, much like individuals' physiological needs (e.g., hunger, thirst) have to be met to grow physically, the satisfaction of the psychological needs is required to foster psychological growth, integrity, and well-being. Because these psychological needs are inherent, it logically follows that they are operative for individuals across the world (i.e., universality claim) and throughout the life span (i.e., cross-age relevance). That is, the satisfaction of these psychological needs represents critical nutrients for the psychological growth of all individuals, regardless of their cultural background, gender, or age (Henning et al., 2019; Nalipay et al., 2020; Niemiec & Ryan, 2009).

SDT's universality argument elicited quite some controversy and enhanced discussion in the field (e.g., Markus & Kitayama, 2003), in part because SDT scholars take a strong position regarding human nature (Deci & Flaste, 1995; Ryan & Deci, 2017; Vansteenkiste & Ryan, 2013). Rather than assuming that people are born as blank slates, SDT assumes that people are equipped with a set of psychological needs that underlie their proactive and growth-oriented nature. That is, people would naturally engage in *need-crafting* behavior (Laporte et al., 2021, 2022), thereby actively seeking out activities and developing relations in which they experience a sense of volition, mastery, and connection. People are receptive for and gravitate toward need-conducive contexts, activities, and relational partners that, in turn, enhance well-being.

Conceptually, the psychological needs have a double function. They serve as instigators of action, thereby determining the direction of individuals' behavior and leading them to proactively shape their environment (see Legault et al., 2017). A key illustration of students' proactivity in the classroom is their agentic engagement (Reeve, 2013). Agentic students voice their preferences, interests, and problems in class, thereby providing opportunities to teachers to attune their teaching to the students' perspective (Matos et al., 2018; Reeve, 2013). As a result, agentic students have a greater likelihood of getting their psychological needs met, which fosters continued proactivity (Patall et al., 2019).

At the same time, need satisfactions are considered critical inputs for students' high-quality motivation, learning, and well-being (Ryan & Deci, 2020). That is, the satisfaction of these needs serves as an energetic resource in general and critical motivational nutrient in particular. Thus, the notion of psychological needs sheds light on both the direction and the energetic basis of motivated action, two aspects each motivation theory needs to address (Deci & Ryan, 1985). In the next section, we address the question how need satisfaction and its contextual support spurs intrinsic motivation.

Energizing Intrinsic Motivation: The Role of Situational Need Support

The introduction of these psychological needs helps to explain under which circumstances individuals' intrinsic motivation gets spurred and when it plummets (Deci & Ryan, 2000). To the extent learners' psychological needs are supported, they will develop a stronger interest in the learning activity at hand, while the thwarting of their needs comes at the cost of their learning enjoyment. Dozens of experimental studies have provided evidence for the contextual factors that trigger situational intrinsic motivation, that is, individuals' interest in and enjoyment of the activity at hand (Vallerand, 1997).

To illustrate the importance of competence support, elementary school children who received normative, positive feedback after solving a series of puzzles reported greater interest and choose to continue working at harder puzzles during a free-choice period compared to those who received negative, normative feedback (Mabbe et al., 2018). Elevated competence need satisfaction could explain the higher intrinsic motivation and challenge seeking in the positive feedback condition. In contrast, normative negative feedback undermines individuals' need for competence, with the resulting perception of failure taking away a lot of the interest in the activity at hand (see Fong et al., 2019 for a meta-analysis and more detailed discussion). Apart from the type of provided feedback, also the nature of the offered learning activity itself can influence

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learners' intrinsic motivation. To illustrate, elementary school children who solved a difficult set of math exercises reported less interest and sought less challenge compared to those who received a relatively easier set of math exercises, an effect that could be accounted for by reduced competence satisfaction (Baten et al., 2020).

Other external events, such as the offer of choice, foster intrinsic motivation through supporting learners' autonomy (e.g., DeMeester et al., 2020; Jang et al., 2016). For instance, elementary school children who got the chance to engage in a preferred painting activity reported greater interest and vitality compared to those who were deprived of their preferred choice, an effect that could be accounted by higher subjective experience of having choice (Waterschoot et al., 2019). Similarly, the experimental variation in the use of inviting and informational, relative to forceful and pressuring, language when introducing learning tasks (Vansteenkiste et al., 2004) and syllabi in higher education (Young-Jones et al., 2019) or when communicating feedback (Mabbe et al., 2018), was found to affect learners' interest through their experiences of autonomy. Further, such inviting language dampened the competence- and autonomy-frustrating effect of offering difficult, relative to medium-easy math exercises, indicating that alterations in communication style can affect perceived hardiness of school task (Baten et al., 2020; see also Patall et al., 2018). While a portion of these experimental studies isolated autonomy—and competence-supportive components (e.g., offering choice; positive feedback; using inviting language; De Muynck et al., 2017; Schneider et al., 2018) to examine its unique potential to uplift students' intrinsic motivation, other studies have simultaneously introduced different need-supportive components (e.g., Coudeville et al., 2020).

When learners feel seduced or manipulated to partake in an already interesting activity, their interest in the activity fails to develop or even plummets all together. To illustrate, monetary rewards obstruct individuals' intrinsic motivation if they are perceived as evaluative (Deci et al., 1999). When introduced in a controlling way, learners' locus of causality (deCharms, 1968) shifts from an internal to an external one, thereby diminishing their need for autonomy and forestalling their intrinsic motivation (Houlfort et al., 2002). Experimental evidence for the differential motivating impact of rewards on intrinsic motivation as a function of their attributed meaning was recently provided (Landry et al., 2020). When monetary rewards were introduced in a controlling way ("the monetary reward is used as a mean to reinforce the performance standards for the task"), they were found to undermine participants' interest in a word scrambling task compared to when the same monetary reward was introduced in an informational way ("the monetary reward is introduced as a token of appreciation for your contribution"; Landry et al., 2020).

Whereas autonomy and competence are integral to students' intrinsic motivation (Reeve & Sickenius, 1994), relatedness may play a more distant role (Deci & Ryan, 2000; Vansteenkiste, et al., 2010). This is because the presence of others is not a necessary prerequisite for the engagement in intrinsically motivated activities (e.g., reading a book, doing homework). Yet, if students experience a strong bond with their peers or teacher, they may enjoy going to school more and report higher school well-being (Leon & Liew, 2017), although they may not by definition find the content of their schoolwork to be more interesting. Yet, if learners can share their learning interests with like-minded peers, they may develop a greater interest in the learning material.

The current trend of integrating technology in education provides new avenues to foster psychological need satisfaction and resulting intrinsic motivation. To illustrate, there is preliminary evidence that a flipped classroom approach in higher education fosters greater need satisfaction and interest in the learning material (Zainuddin & Pererar, 2019). As is the case for any external event (e.g., rewards, feedback, surveillance), the question whether innovative pedagogical and technological methods foster greater intrinsic motivation depends on whether their need-enabling potential is fully actualized (e.g., Jeno et al., 2019). To the extent that greater opportunities for initiative and voice (autonomy), skill acquisition and progress-oriented feedback (competence), and warm peer interactions (relatedness) are built into these new methods and technologies, they may help to maintain and even foster learners' interests. Yet, if poorly structured or not aligned with learners' skill levels, learners may also feel left by themselves such that their psychological needs get met less, if not thwarted. Thus, the angle of psychological needs provides scholars and educators with a critical criterion to design, evaluate, and optimize any innovative technological and pedagogical methods to promote intrinsic motivation in learners.

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Anchoring Intrinsic Motivation: The Role of Enduring Need Support

Supporting learners' need satisfaction during a specific learning activity is a first critical step to foster interest. Yet, to foster sustainable intrinsic motivation, caregivers need to *repeatedly* create a need-supportive climate. Dozens of studies have confirmed the crucial role of both teacher and parental need support for learners' intrinsic motivation to learn, both on a day-to-day basis (Patall et al., 2018) and over time (Gottfried et al., 2009). Such ongoing support of learners' psychological needs helps them to transform their experienced intrinsic motivation for a given task, which could be fleeting and short-lived, into a more enduring interest that guides subsequent behavior. The *developmental ideal* then is that unfolding interests become more stable over time, while getting gradually refined and anchored within learners' identity (i.e., a harmonious passion).

Children are more likely to develop such a mature and full-fletched intrinsically motivated (academic) identity when caregivers provide sufficient room for their offspring to explore and experiment with different emerging interests, without pressuring them into a particular direction or pushing them toward early closure of available options. Choice in combination with caregivers' patience to follow the rhythm of the child's exploration are key factors for the development of long-term, enduring intrinsic motivations. Also, to maintain and develop children's inclination to engage in spontaneously attractive activities, they need to experience sufficient competence support and interest from their caregivers. Through offering challenge and caregivers' shared engagement in activities that fit with children's "raw" preferences, they can build a sense of confidence and value their engagement as such.

Further, for this development to take place, caregivers also do well to foster children's inner valuing, thereby actively encouraging their children to reflect about their most important interests and preferences when making choices in life (e.g., hobbies and study choices) (Assor et al., 2020). Such introspection allows learners realize more clearly which activities or roles are truly fascinating or appealing to them (or, conversely, which activities and roles they would adopt only to gain social approval or to meet external expectations), thereby enhancing their *motivational awareness.* That is, children learn to differentiate between the different motives that underlie their behavior, thereby discriminating better between what they find truly interesting and enjoyable and what is more externally driven (Guay & Bureau, 2018). Such motivational differentiation then allows them to make informed and more authentic identity-relevant choices.

In short, against the backdrop of a need-supportive family climate, children are better capable of developing a mature (i.e., enduring, refined, and well-anchored) intrinsically motivated identity, which helps them to avoid the pitfalls of indecisiveness when they eventually need to make career choices (Guay et al., 2006; Skhirtladze et al, 2019). Of course, during the choice process itself and the subsequent transition phase (Duineveld et al., 2017), learners need to experience their caregivers as need-supportive such that they eventually make a career choices that is grounded in these intrinsically motivated identities. If this is the case, these well-anchored career choices will feed into learners' intrinsic motivation for a specific course or task (Yu et al., 2018). Due to their more agentic engagement, intrinsically motivated learners may then elicit a need-supportive interaction pattern from teachers (Cuevas et al., 2018), which reinforces learners' intrinsically motivated career choices, thus leading to a positive spiral (Matos et al., 2018).

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Psychological Needs as Universal Nutrients of Engagement, Learning, and Well-Being

Psychological need satisfaction not only serves to energize intrinsic motivation, but has been found to predict a broad array of educational and well-being benefits. To illustrate, in samples of Korean and American students, Jang et al. (2009) examined to what extent each of these three psychological needs underlies both students' most and least satisfying experiences at school. After having recalled an (un)satisfying event at school, high school students rated the extent to which eight different candidate needs (i.e., self-actualization, stimulation, safety-security; self-esteem; popularity-influence), including the SDT trio, were fulfilled during the recalled event. Especially autonomy and competence were most salient during both types of self-generated events and accounted for substantial variation in participants' event-related affect. Further, teachers seem to be aware of the crucial role of these psychological needs in children's healthy development. Specifically, when being asked to rate a list of 26 possible needs in terms of their importance for children's development as a psychologically healthy, successful and harmonious person (Lynch & Salikhova, 2016), Russian teachers listed competence and relatedness as top nutrients. Autonomy, in contrast, was valued much less, which the authors interpreted as being a function of the different cultural circumstances.

The number of studies providing evidence for the resourceful role of psychological need satisfaction has grown exponentially over the past decade. Students who experience greater need satisfaction report greater engagement (e.g., Koch et al., 2017; Oga-Baldwin et al., 2017), more deep-level learning (e.g., Betoret & Artiga, 2011), and are more satisfied with themselves (e.g., Standage & Gillison, 2007) and with their lives in general (e.g., Véronneau et al., 2005). The benefits of need satisfaction were also evidenced when teacher ratings of school adjustment (Ahmad et al., 2013) or behavioral engagement (Standage et al., 2006) were used.

There is also longitudinal evidence for the academic benefits associated with need satisfaction. Surveying eighth grade students multiple times during an entire semester, Jang et al. (2012) reported that increased need satisfaction halfway through the semester related to improved engagement by the end of the semester, which, in turn, predicted higher achievement. In the most encompassing longitudinal study to date, spanning a six-year period from the end of elementary school to the of high school, Ratelle et al. (2007) found considerable heterogeneity in students' need-related developmental trajectories. Students in trajectories of either stable high or increasing need satisfaction reported higher levels of academic, social, and emotional adjustment at the end of high school.

Further speaking to the universal importance of need-based experiences is the observation that their beneficial role has been demonstrated across different age groups, including elementary, middle, high school, and university students (e.g., Rodriguez-Meirinhos et al., 2020; Tian et al., 2016); across courses, including reading (e.g., De Naeghel et al. 2012), science (e.g., Patall et al., 2016), music (e.g., Evans & Bonneville-Roussy, 2016) and physical education (e.g., Taylor & Lonsdale, 2010) to name a few; at both the between-person and within-person level, with students' need satisfactions fluctuating from week to week (Campbell et al., 2018), from day-to-day (Van der Kaap-Deeder et al., 2017), and even from lesson to lesson (Krijgsman et al., 2017); among both non-gifted and teacher-nominated gifted students (Hornstra et al., 2020); and in both face to face as well as online learning contexts (Wang et al., 2019).

Although mean-level differences in need-based experiences do emerge as a function of these different characteristics, the more general point within BPNT is that the functional benefits associated with need satisfaction are universal. This position still leaves room for nuance and variation, with both the strength of the benefits associated with need satisfaction and the pathways leading to need satisfaction being somewhat sample- and situation-dependent (Ryan et al., 2019; Vansteenkiste et al., 2020). That is, the universality claim does not imply perfect uniformity, but allows some variation in the functional effects of and contexts fostering need satisfaction (Soenens et al., 2011).

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When the Learning Material Is Boring or Hard: The Role of Internalized Motivation

Apart from fostering intrinsic motivation, need satisfaction plays a critical role in the process of internalization. We first describe different regulatory subtypes that vary in their degree of internalization. After discussing how the internalization continuum provides a more refined insight in learners' motivational functioning, we review a number of critical considerations. We conclude the chapter with a discussion of the critical environmental components that are conducive to the process of internalization.

Need Satisfaction Fosters Internalization

Of course, not all children find their schoolwork interesting, on the contrary. Some children may simply find it boring and studying thus requires considerable effort from them. Moreover, even children who are generally intrinsically motivated for school encounter moments of boredom and are confronted occasionally with unappealing learning materials. To foster greater ownership around non-interesting or even boring and tough schoolwork, the same three psychological needs are said to be critical. That is, students will begin internalizing the personal importance of their studying, thereby perceiving their schoolwork as increasingly *self-relevant* when they experience greater need satisfactions (Ryan & Deci, 2017; Vansteenkiste et al., 2018). Competence and relatedness satisfaction are considered essential ingredients for individuals to begin taking in the reason for performing the assigned learning material, yet, for full internalization to occur they also need to experience a sense of volition and psychological freedom (i.e., autonomy satisfaction). That is, if learners experience a strong bond with their teacher and feel effective in carrying out the activity, they may merely do so to please their teachers, to gain the teachers' approval, or to avoid feeling guilty for being disloyal (Calvo et al., 2010; Haerens et al., 2015). Although the regulation has been taken in, the activity is not performed wholeheartedly. For learners to fully endorse an activity, they must have the feeling that they took personal responsibility for the assigned learning activity (Markland & Tobin, 2010; Vansteenkiste et al., 2018).

As articulated within *Organismic Integration Theory* (Ryan & Deci, 2017; Vansteenkiste et al., 2018), depending on the level of experienced need satisfaction, learners will either partially or more fully internalize the regulation underlying their learning behavior. *Internalization denotes the process through which externally offered guidelines, requests, and norms have been transformed into personally endorsed and, hence, volitionally adhered guidelines, requests, and norms.* Figure 1 provides an overview of the different types of regulation.

Figure 1:

Type of Motivation	Extrinsic motivation				Intrinsic motivation
Type of Regulation	External regulation	Introjected regulation	Identified regulation	Integrated regulation	Intrinsic regulation
Motivational force	Commands, rewards, punishments	Guilt, shame, ego-involvement	Personal significance and value, relevance	Harmony and anchoring within values, commitments	Interest, enjoyment, curiosity
Internalization	Lack of internalisation	Partial	Full	Fullest	Not required

Overview of Different Motivational Subtypes (adapted from Ryan & Deci, 2017)

In the case of external regulation, learners simply engage in the learning to obtain contingent rewards and approval or avoid punishments and criticism. Because the reason for performing the activity is completely outside themselves, the behavioral regulation has not been taken in at all. In the case of introjected regulation, the learning is buttressed with feelings of guilt, shame, and contingent self-worth such that learners now feel pressured from "within" instead of "without" to put effort in their learning (Assor et al., 2009). Yet, because the reason for performing the activity is not fully accepted and because the activity engagement is associated with feelings of ambivalence and internal conflict (Kelly et al., 2015), introjected regulation only represents partial internalization.

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A fuller form of internalization is achieved when learners see the personal value of the learning activity (*identified regulation*) or when the learning is conceived to be congruent with deeply held values, commitments, and preferences (*integrated regulation*). In the latter case, the learning activity is deeply anchored and thus fully internalized, thereby forming a solid basis for durable engagement in the activity. Apart from this vertical aspect of integrated regulation, learners can also ask themselves whether their learning is (horizontally) compatible with activities in other important life domains (e.g., leisure time, household). Children and adolescents wear many hats, with each hat representing a different identity that can be more or less in harmony with other identities. Full internalization thus also manifests when different identities reinforce one another instead of being conflicting (Ryan & Deci, 2003; Soenens & Vansteenkiste, 2011).

This process of internalization is not needed when learners find the learning material interesting to begin with. When intrinsic motivation is present, learners already willingly engage in the learning activity. As such, the reason for their activity engagement does not need to be taken in. Therefore, intrinsic motivation is technically speaking not part of the internalization continuum but rather represents the most self-determined form of motivation on the more encompassing self-determination continuum. Because identified and integrated regulation as well as intrinsic motivation are all characterized by volitional properties, these motivation subtypes have sometimes been grouped under the label of autonomous motivation. Autonomous motivation is then contrasted with controlled motivation, which represents the combination of external and introjected regulation, the two more pressured forms of regulation (Ryan & Deci, 2020).

Refined Insight in Learners' Motivational Functioning

The internalization continuum allows for a more refined view on learners' extrinsic motivation. Rather than treating the different extrinsic motives as a homogeneous category, subtypes are distinguished that vary in their level of ownership and self-determination. Research convincingly demonstrated that the correlates with desirable learning outcomes (e.g., self-regulated learning, engagement, well-being) become increasingly positive as one moves along the continuum from external to more internalized forms of motivation (Sheldon et al., 2017). Similarly, the pattern of correlates with undesirable outcomes (e.g., disengagement) becomes increasingly negative as one moves along the continuum (Howard et al., 2017).

This gradual pattern of correlates does not preclude the possibility that the peak in the observed pattern of correlations may—at least in some cases—be outcome-dependent. For instance, especially introjected regulation relates to learner's test anxiety as learner's ego is more at stake and thus potentially threatened by poor exam results when introjection is a salient motive (Ryan & Connell, 1989; Van der Kaap-Deeder et al., 2016). When predicting learners' diligence and subsequent performance for harder and non-interesting tasks, identified regulation is a more critical predictor than intrinsic motivation (Burton et al., 2006). Presumably, identified regulation and a strong personal conviction are needed for students to stay determined in the face of harder or non-interesting parts of the learning material. Thus, when predicting particular outcomes, both the location of the subtype on internalization continuum and the specific psychological dynamics associated with each subtype need be taken into account.

Whereas the motivational subtypes have been studied in isolation in some studies, other studies have relied on composite scores, thereby mainly contrasting controlled and autonomous forms of motivation. The more students report autonomous motivation for their schoolwork in general (Ryan & Connell, 1989) or in relation to course-specific subjects (e.g., Guay & Bureau, 2018), the more they benefit in turn, as indexed by improved engagement, well-being, and performance (e.g., De Naeghel et al., 2012).

Interestingly, autonomous motivation is related to higher achievement even when controlling for learners' intelligence (Boncquet et al., 2020). In a sample of elementary school children transitioning to secondary school, students with higher autonomous motivation scored better on standardized achievement math tests overall and also made greater progress in their achievement across the transition over and above effects of intelligence.

A variety of mechanisms may explain the achievement benefits associated with autonomous motivation. Because autonomously motivated students act upon their personally held values, interests, and preferences, they have more energy available to deeply process the learning material, signaling a higher cognitive engagement (Vansteenkiste et al., 2005). Also, autonomous motivation has been found to predict better

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planning and time management and less procrastination such that autonomously motivated students are better prepared for tests and less likely to be overwhelmed by paralyzing test anxiety (e.g., Vansteenkiste et al. 2009). Finally, because interest- and value-based behavior forms a solid basis to act from, autonomously motivated students act in more resilient and less defensive ways when they encounter stressors (Ntoumanis et al., 2009; Weinstein & Ryan, 2011). That is, autonomously motivated students may both appraise the encountered obstacles as less stressful and threatening and use more adaptive coping strategies in handling the encountered stressor, which helps explain their better performance (Amiot et al., 2008; Bonneville-Roussy et al., 2017).

Critical Considerations

The self-determination continuum typically elicits a number of reflections in readers that we aim to address in this last section. Readers may raise the following issues: (a) "Do purely intrinsically motivated students really exist?" and (b) "Are these different motives operative regardless of students' ages?"

Pure Intrinsic Motivation?

Purely intrinsically motivated learners may be rare, especially in schools where evaluations, deadlines, and other need-threatening external events are operative. Also, theoretically, there is no reason to expect that learners should be exclusively intrinsically motivated. In fact, the self-determination continuum suggests that a variety of reasons can underlie students' functioning as our behavior is *multi-determined*. Yet, the precise combination of motives that characterize learners may vary, with different combinations representing different motivational profiles. Whereas some leaners may combine intrinsic and identified reasons, thus being guided by both interest and value, both motives may be disconnected or absent among other learners. For instance, dutiful learners may combine high identified and introjected regulation, a combination that may not necessarily lead to greater enjoyment of students' learning. A rapidly increasing number of studies have adopted a person-centered perspective to shed light on the presence and stability vs. change of these different motivational profiles (e.g., Vansteenkiste et al., 2009; Corpus et al., 2009). Mirroring the effects at the dimensional level, these studies have shown that motivational profiles characterized by the stronger presence of autonomous and reduced presence of controlled motives display the most adaptive pattern of functioning.

Two other issues deserve being highlighted. First, although there exist inter-student differences in intrinsic motivation, a student's interest may also undergo ups and downs from week to week (Vandenkerckhove et al., 2019), from day to day (Patall et al., 2018), or even from lesson to lesson (Tsai et al., 2008). Second, some behaviors at school, like adhering to class- and school-based rules, may not be intrinsically motivating at all (Aelterman et al., 2019). So, it is especially critical that learners come to endorse and internalize these rules such that they take more personal responsibility for themselves, their peers, and their school environment. Indeed, more identified regulation related to less resentment, acting out, and truancy, even when controlling for students' study motivation (Aelterman et al., 2019).

Age Specificity?

Another question is whether the various motives on the self-determination continuum are applicable to different age groups. These various motivations can be operative at a younger age, yet children may not necessarily be aware of the motives driving their behavior. Because children have not developed yet the capacity for introspection and meta-cognition, these motives may be primarily observable through behavior. To illustrate, toddlers' exploration and discovery and their natural tendency to act prosocially (Warneken & Tomasello, 2008) signal the presence of natural growth process. More empirical work in this area would be welcomed, for instance by shedding light on the question how the different motives on the continuum get gradually differentiated across age (Guay et al., 2018).

Depending on age and need-supportive circumstances, children may develop an increasing susceptibility for and refined awareness of the motives that guide their behavior. Especially integrative regulation may emerge only at a later age and even only under specific circumstances, for instance, when students are making identity-relevant decisions. To illustrate, when making a career decision, last year high school students could wonder whether their college studies fit with other identities, such as being a soccer player or being a leader at a youth movement. To the extent that these different identities get harmonized,

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students are likely to report greater integrated regulation during their actual studies. Although reflective capacities and increasing awareness are key to move toward greater ownership, there is also a more experiential in-road to greater ownership that may be more applicable for young children. Take the example of an 18-month-old toddler being on the point of running to a fireplace. While parents may prevent the child from burning its hand by explaining her that fires are hot, they could as well take her to the fireplace and ask her to experience herself what hot means. Although the child has not yet the cognitive capacities to understand the explanation, experience-based learning may lead the child to move quickly across the internalization continuum toward higher ownership.

Fostering Internalization: The Role of Contextual Need Support

Similar to the critical role of contextual need support to promote intrinsic motivation, the nurturance of learners' psychological needs is also key to promote greater internalization and ownership. Need-supportive socialization matters both in a specific situation, leading to greater endorsement of learning activities at hand, as well as in a more enduring way, thereby contributing to the development of a more strongly anchored set of values, commitments, and aspirations over time (Vansteenkiste et al., 2018). As is the case for the growth process of intrinsic motivation, dozens of experimental and correlational studies have sought to identify the key contextual features that foster internalization. The starting point to support the process of internalization (relative to intrinsic motivation) is somewhat different as it is often harder to connect to students who find little interest in the learning content or even dislike assigned learning activities all together. It is therefore critical to empathize with the learners' perspective and to explore the obstacles for initiating or persisting at the learning activity. Dependent upon the encountered roadblocks, teachers can roll with learners' resistance and felt irritation, provide a meaningful rationale for the requested learning tasks, or provide additional guidance and feedback to overcome the obstacles (Aelterman et al., 2019; Reeve, 2009).

While some experimental studies have isolated and tested the unique effect of different need-supportive strategies (e.g., Jang, 2008), others have made use of a more encompassing approach, thereby simultaneously operationalizing different key components (e.g., Deci et al., 1994 Pelletier & Joussemet, 2017). To illustrate, the provision of a meaningful rationale for a statistics lesson was found as a buffer against learners' decline in observed engagement, an effect accounted for by higher internalization of the learning activity. If such rationales are coupled with other autonomy-supportive strategies, such as acknowledging negative affect, they were found to yield a more pronounced effect (Steingut et al., 2017). Interestingly, the benefits of a combined autonomy-supportive approach have been documented in non-normative populations, including individuals' mild intellectual disabilities (Pelletier & Joussemet, 2017) and youngsters with emotional-behavioral problems (Savard et al., 2013) who engaged in a tedious and unpleasant task.

When learners lack interest and act in oppositional ways, it is very challenging to remain need-supportive. Disengaged learners often pull a more controlling approach, with socializing agents justifying their use of pressure by arguing that it is a stepping stone toward increasing ownership. Pressure would be the necessary springboard for more adaptive subtypes of motivation to emerge over time. This reasoning is grounded in the (implicit) assumption that the different type of regulation on the self-determination continuum represent different phases learners need to cycle through. Yet, Organismic Integration Theory is not a phase-like developmental model, which would imply that students progress linearly through the different subtypes to move toward the desired endpoint of autonomous motivation. Progress is not necessarily linear but can occur fairly abrupt, with students displaying different motivational trajectories (characterized not only by progress but also by stability and regression) and with psychological need satisfaction playing an important role in students' motivational shifts. So, what disengaged students need is not more pressure but more need-supportive teachers.

When students are put under pressure, their psychological needs and especially the need for autonomy gets frustrated. Although such pressure may lead students to engage in the requested behavior, the behavior is unlikely to be owned and, hence, may fade again when the pressures no longer apply. Longitudinal studies do not provide evidence that controlled motivational subtypes predict increases in autonomous subtypes that are located higher up on the continuum. On the contrary, Taylor et al. (2014) found that external regulation was negatively related to students' intrinsic motivation one year later. Rather than subjecting disinterested students to pressure, they benefit from a need-supportive teacher who is willing to empathize

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with their perspective, to explore their obstacles, to provide a meaningful rationale for the requested learning tasks, and to provide additional guidance and feedback to overcome the obstacles (Aelterman et al., 2019; Reeve, 2009).

When students' lack of motivation for school is chronic, students may also want to reconsider their study choice and choose a different direction that better matches their interests, preference, and competence level. Such a reorientation provides them an opportunity to become better aware of their emerging ideals, interests, and commitments and to act upon them. As pointed out by Assor and colleagues (Assor, 2018), on these critical decision moments socializing figures need to encourage and support youngsters in examining their values, interests, and commitments and to foster the inner valuing of these inner resources. Such need-supportive practices allow youngsters to better found and anchor their career choices, with resulting positive effects on their study motivation (Assor et al., 2020).

Conclusion

The majority of, if not all, teachers would agree that it is a pleasure and their strong wish to teach and interact with motivated students. While teachers often refer to student motivation in quantitative terms, what is especially critical is that their students display high-quality motivation. Such high-quality motivation manifests through learners' intrinsic and internalized motives for learning. When learners find interest in the learning material and see its relevance and value, they are more likely to deeply process it, persist, and report higher well-being. Although children are naturally attracted to follow their interest and have a readiness to assimilate and accept social norms, these growth processes need to be energized through the satisfaction of learners' needs for autonomy, competence, and relatedness. Social environments conducive to learners' needs promote greater intrinsic motivation for and ownership of the activity at hand, while also allowing learners to develop a more refined, well-anchored, and harmonized set of values, interests, and preferences, which serve as a more stable basis for regulating their day-to-day learning activities.

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